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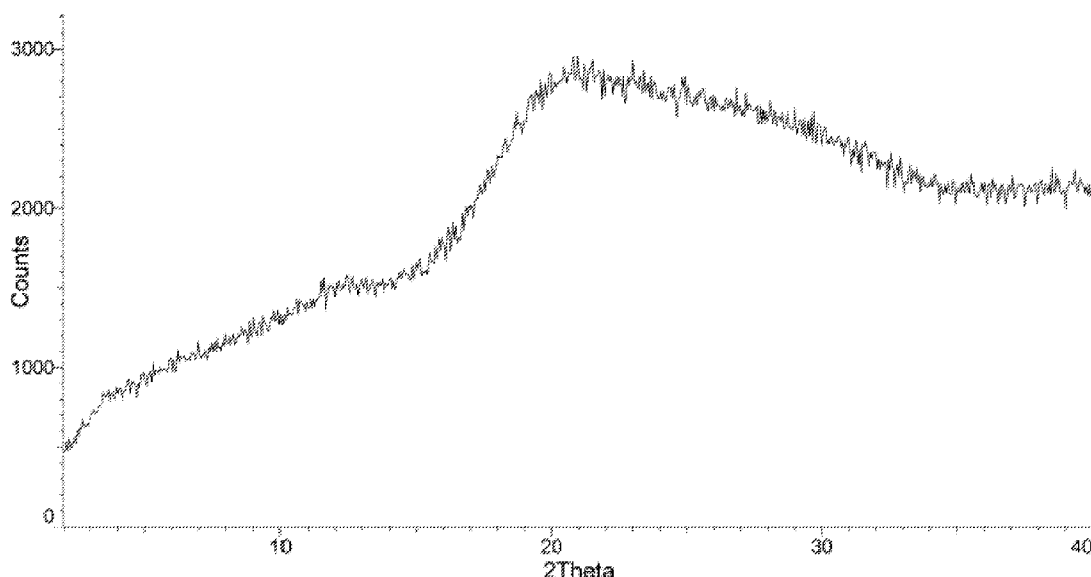
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(54) Title: SOLID STATE FORMS OF LOTILANER AND PROCESS FOR PREPARATION THEREOF

Figure 1: A characteristic X-ray powder diffraction pattern (XRPD) of amorphous Lotilaner



(57) Abstract: The present disclosure encompasses a solid-state form of Lotilaner, in embodiment processes for preparation thereof, and pharmaceutical compositions thereof. The present disclosure further encompasses Lotilaner salts and their solid state forms, as well as processes for preparation thereof, and pharmaceutical compositions thereof.

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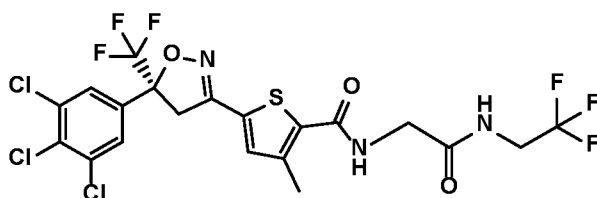
SOLID STATE FORMS OF LOTILANER AND PROCESS FOR PREPARATION THEREOF

FIELD OF THE DISCLOSURE

[0001] The present disclosure encompasses solid state forms of Lotilaner, in embodiments processes for preparation thereof, and pharmaceutical compositions thereof. The present disclosure further encompasses Lotilaner salts and their solid state forms, as well as processes for preparation thereof, and pharmaceutical compositions thereof.

BACKGROUND OF THE DISCLOSURE

[0002] Lotilaner, chemical name 3-Methyl-*N*-[2-oxo-2-(2,2,2-trifluoroethylamino)ethyl]-5-[(5*S*)-5-(3,4,5-trichlorophenyl)-5-(trifluoromethyl)-4*H*-1,2-oxazol-3-yl]thiophene-2-carboxamide, has the following chemical structure:



[0003] Lotilaner is a veterinary drug administered orally for the treatment and prevention of flea and tick infestations in dogs. Lotilaner has also been investigated for use in the killing of ticks attached to human skin. It is currently developed for the treatment of eye infection and blepharitis.

[0004] The compound is described in International Publication No. WO 2010/070068.

[0005] Polymorphism, the occurrence of different crystalline forms, is a property of some molecules and molecular complexes. A single molecule may give rise to a variety of polymorphs having distinct crystal structures and physical properties like melting point, thermal behaviors (e.g., measured by thermogravimetric analysis (“TGA”), or differential scanning calorimetry (“DSC”)), X-ray diffraction (XRD) pattern, infrared absorption fingerprint, and solid state (¹³C) NMR spectrum. One or more of these techniques may be used to distinguish different polymorphic forms of a compound.

[0006] Different salts and solid state forms (including solvated forms) of an active pharmaceutical ingredient may possess different properties. Such variations in the properties of different salts and solid state forms and solvates may provide a basis for improving formulation,

for example, by facilitating better processing or handling characteristics, changing the dissolution profile in a favorable direction, or improving stability (polymorph as well as chemical stability) and shelf-life. These variations in the properties of different salts and solid state forms may also offer improvements to the final dosage form, for instance, if they serve to improve bioavailability. Different salts and solid state forms and solvates of an active pharmaceutical ingredient may also give rise to a variety of polymorphs or crystalline forms, which may in turn provide additional opportunities to assess variations in the properties and characteristics of a solid active pharmaceutical ingredient.

[0007] Discovering new solid state forms and solvates of a pharmaceutical product may yield materials having desirable processing properties, such as ease of handling, ease of processing, storage stability, and ease of purification or as desirable intermediate crystal forms that facilitate conversion to other polymorphic forms. New solid state forms of a pharmaceutically useful compound can also provide an opportunity to improve the performance characteristics of a pharmaceutical product. It enlarges the repertoire of materials that a formulation scientist has available for formulation optimization, for example by providing a product with different properties, including a different crystal habit, higher crystallinity, or polymorphic stability, which may offer better processing or handling characteristics, improved dissolution profile, or improved shelf-life (chemical/physical stability). For at least these reasons, there is a need for additional solid state forms (including solvated forms) of Lotilaner.

SUMMARY OF THE DISCLOSURE

[0008] The present disclosure provides solid state forms, particularly crystalline polymorphs, of Lotilaner, processes for preparation thereof, and pharmaceutical compositions thereof. The present disclosure further encompasses Lotilaner salts and their solid state forms, as well as processes for preparation thereof, and pharmaceutical compositions thereof.

[0009] These solid state forms, particularly crystalline polymorphs of Lotilaner or Lotilaner salts can be used to prepare other solid state forms of Lotilaner, Lotilaner salts or co-crystals and their solid state forms.

[0010] The present disclosure also provides uses of the said solid state form of Lotilaner or Lotilaner salts in the preparation of other solid state forms of Lotilaner, Lotilaner salts or co-crystals salts thereof.

- g) a combination of two or more of (a), (b), (c), (d), (e), and (f).
20. Crystalline Lotilaner according to claim 1 or claim 19, designated form LT4, characterized by an X-ray powder diffraction pattern having peaks at 12.4, 14.7, 19.0, 22.2 and 26.7 degrees 2-theta $\pm$ 0.2 degrees 2-theta, and also having any one, two, three, four or five additional peaks selected from 17.9, 20.4, 23.0 and 29.7 degrees 2-theta $\pm$ 0.2 degrees 2-theta.
 21. Crystalline Lotilaner according to claim 1, claim 19 or claim 20, designated form LT4, characterized by an X-ray powder diffraction pattern having peaks at 12.4, 14.7, 17.9, 19.0, 20.4, 22.2, 23.0, 26.7 and 29.7 degrees 2-theta $\pm$ 0.2 degrees 2-theta.
 22. Crystalline Lotilaner according to any of Claims 1, 19, 20, or 21, designated form LT4, characterized by a TGA thermogram substantially as depicted in Figure 11, or by a DSC thermogram substantially as depicted in Figure 10, or by combination of these data.
 23. Crystalline Lotilaner according to any of claims 1, 19, 20, 21, or 22, designated form LT4, wherein the crystalline form is isolated.
 24. Crystalline Lotilaner according to any of claims 1, 19, 20, 21, 22, or 23, designated form LT4, wherein the crystalline form is anhydrous form.
 25. Crystalline Lotilaner according to any of claims 1, 19, 20, 21, 22, 23, or 24, designated form LT4, characterized by water content of less than 1% (w/w), as detected for example by KF and/or TGA.
 26. Crystalline Lotilaner according to any of claims 2-25, which contains: no more than about 20%, no more than about 10%, no more than about 5%, no more than about 2%, no more than about 1% or about 0% of any other crystalline forms of Lotilaner.
 27. Crystalline Lotilaner according to any preceding claim, which contains: no more than about 20%, no more than about 10%, no more than about 5%, no more than about 2%, no more than about 1% or about 0% of amorphous Lotilaner.
 28. Crystalline Lotilaner according to any preceding claim, which is chemically pure.
 29. Crystalline Lotilaner according to any preceding claim, which is enantiomerically pure.
 30. A pharmaceutical composition comprising a crystalline Lotilaner according to any of claims 1-29 and at least one pharmaceutically acceptable excipient.

31. Use of a crystalline Lotilaner according to any of claims 1-29 for the preparation of a pharmaceutical composition and/or formulation, preferably wherein the pharmaceutical formulation is an ophthalmic solution or ophthalmic suspension.
32. A process for preparing the pharmaceutical composition according to Claim 30, comprising combining a crystalline Lotilaner according to any of claims 1-29 with at least one pharmaceutically acceptable excipient.
33. A crystalline Lotilaner according to any of claims 1-29, or a pharmaceutical composition according to claim 30, for use as a medicament.
34. A crystalline Lotilaner according to any of claims 1-29, or a pharmaceutical composition according to claim 30, for use in treatment of eye infections and/or blepharitis.
35. A method of treating eye infections and/or blepharitis, comprising administering a therapeutically effective amount of a crystalline Lotilaner according to any of claims 1-29, or a pharmaceutical composition according to claim 30, to a subject in need of the treatment.
36. Use of a crystalline Lotilaner according to any of claims 1-29, in the preparation of another solid state form of Lotilaner, Lotilaner salt or co-crystals and/or solid state form thereof.
37. A process for preparing a solid state form of Lotilaner, Lotilaner salt and/or solid state form thereof, comprising preparing any one or a combination of a crystalline Lotilaner according to any of claims 1-29, and converting it to another a solid state form thereof, or to Lotilaner salt or co-crystals and/or solid state form thereof.
38. A process for preparing crystalline Lotilaner Form LT5 as defined in any of Claims 2-9, comprising slurring Lotilaner, preferably form LT3, in a mixture of isopropanol and n-heptane.
39. A process according to claim 38, wherein the ratio of isopropanol and heptane is: about 3:1 to about 1:3, about 2:1 to about 1:2, about 1.5:1 to about 1:1.5, about 1.2:1 to about 1:1.2, about 1.1:1 to about 1:1.1, or about 1:1.
40. A process according to any of claim 38 or claim 39, wherein Lotilaner, preferably crystalline form LT3, is combined with the mixture of isopropanol and heptane to obtain a slurry, preferably at room temperature.

41. A process according to any of claims 37, 38, 39, or 40, wherein the slurry is heated, preferably to a temperature of about 40°C to about 80°C, about 50°C to about 70°C, about 55°C to about 65°C, or about 60°C.
42. A process according to claim 41, wherein the mixture is maintained at the heated temperature for a period of: about 3 days to about 21 days, about 7 to about 18 days, about 10 to about 16 days, about 12 to about 15 days, or about 14 days.
43. A process for preparing crystalline Lotilaner form LT5 as defined in any of Claims 2-9, comprising crystallization of Lotilaner from isoamyl alcohol.
44. A process according to claim 43, comprising dissolving Lotilaner in isoamyl alcohol, optionally seeding with Form LT5 of Lotilaner, and cooling.
45. A process according to claim 44, wherein the Lotilaner is dissolved in isoamyl alcohol at a temperature of: about 30°C to about 100°C, about 45°C to about 80°C, about 50°C to about 70°C, about 55°C to about 65°C, or about 60°C.
46. A process according to any of claims 43, 44 or 45, comprising adding seeds of crystalline form LT5.
47. A process according to any of claims 46, wherein the solution is cooled, preferably to a temperature of: about 30°C to about 50°C, about 35°C to about 45°C, or about 40°C prior to adding the seed crystals.
48. A process according to claim 47, wherein the mixture is stirred for a sufficient time to allow crystals of Form LT5 to form, preferably: about 15 minutes to about 4 hours, about 30 minutes to about 3 hours, about 45 minutes to about 2 hours, or about 1 hour.
49. A process according to claim 47 or claim 48, wherein the mixture is further cooled to a temperature of: about 20°C to about 40°C, about 25°C to about 35°C, or about 30°C.
50. A process according to claim 49, wherein the mixture is stirred at this temperature for: about 15 minutes to about 4 hours, about 30 minutes to about 3 hours, about 45 minutes to about 2 hours, or about 1 hour.
51. A process according to any of claims 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, or 50, wherein the obtained solid is isolated, preferably by filtration.
52. A process according to claim 51, wherein the solid is dried, preferably under vacuum, and preferably at a temperature of about 25°C-30°C, preferably for: about 15 minutes to about

4 hours, about 30 minutes to about 3 hours, about 45 minutes to about 2 hours, or about 1 hour.

53. A process according to any of claims 38-52, further comprising combining the Lotilaner with at least one pharmaceutically acceptable excipient to form a pharmaceutical composition.

Figure 1: A characteristic X-ray powder diffraction pattern (XRPD) of amorphous Lotilaner

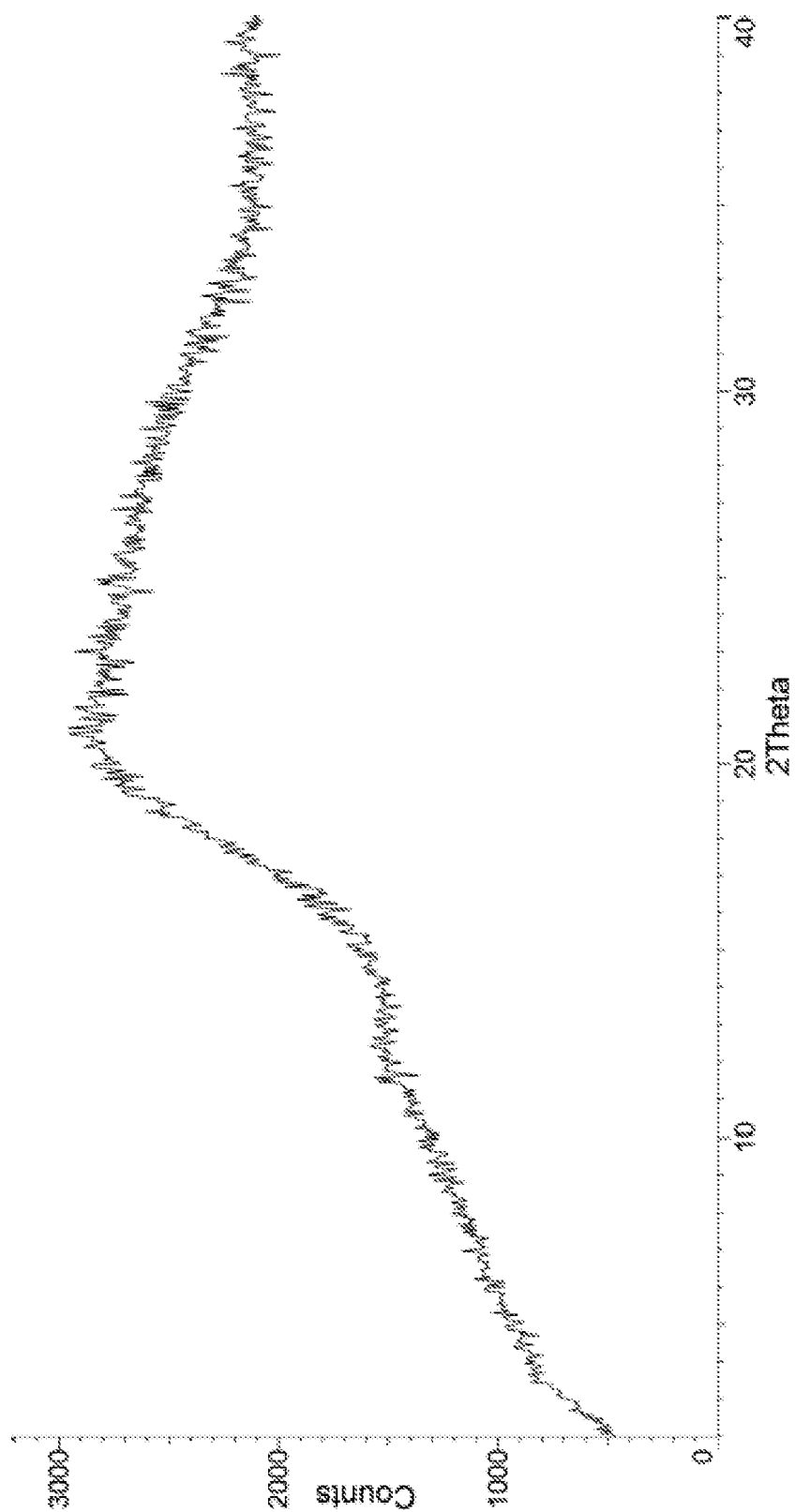


Figure 2: A characteristic XRPD of crystalline Lotilaner Form LT1

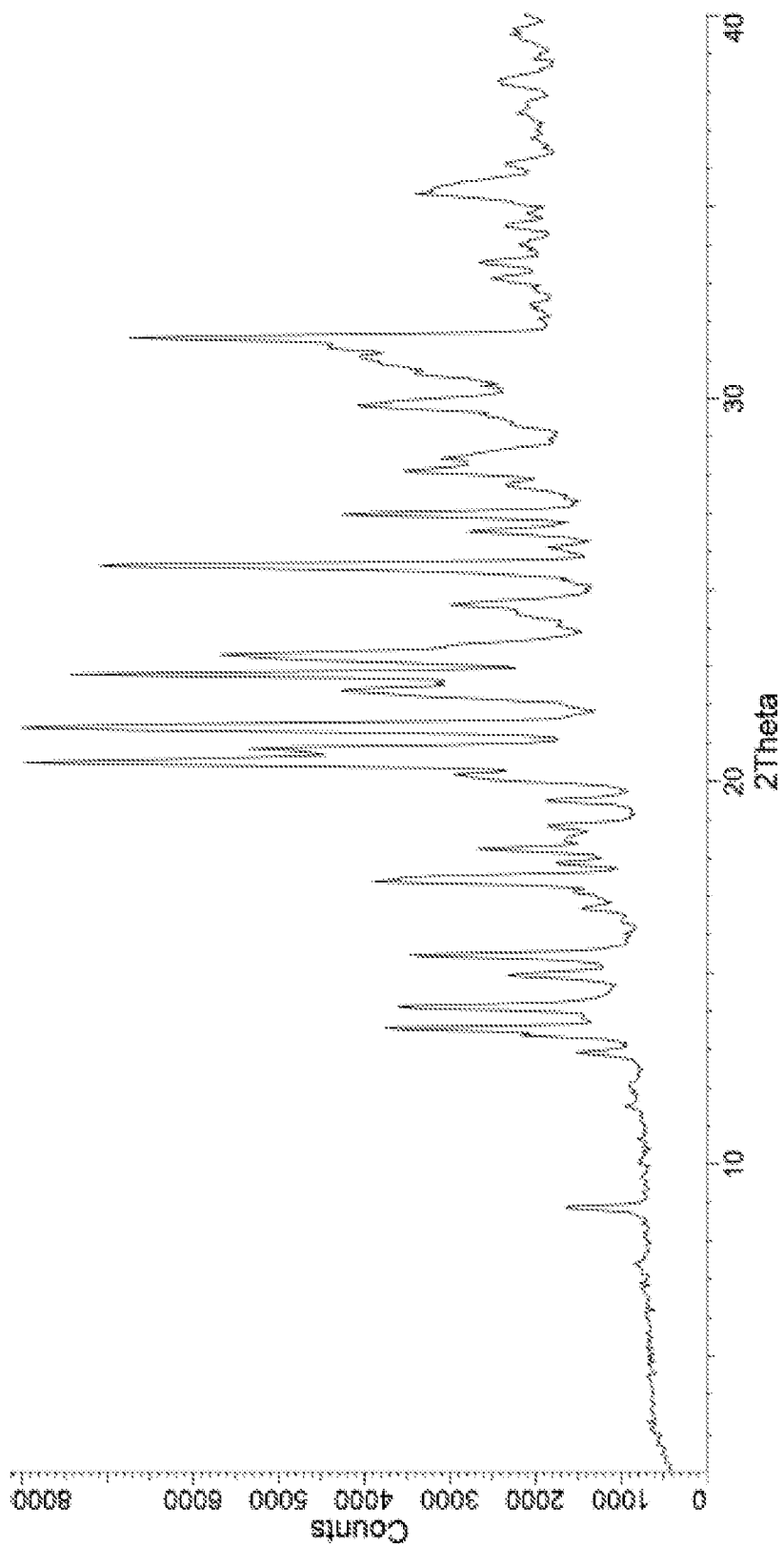


Figure 3: A characteristic XRPD of crystalline Lotilaner Form LT2

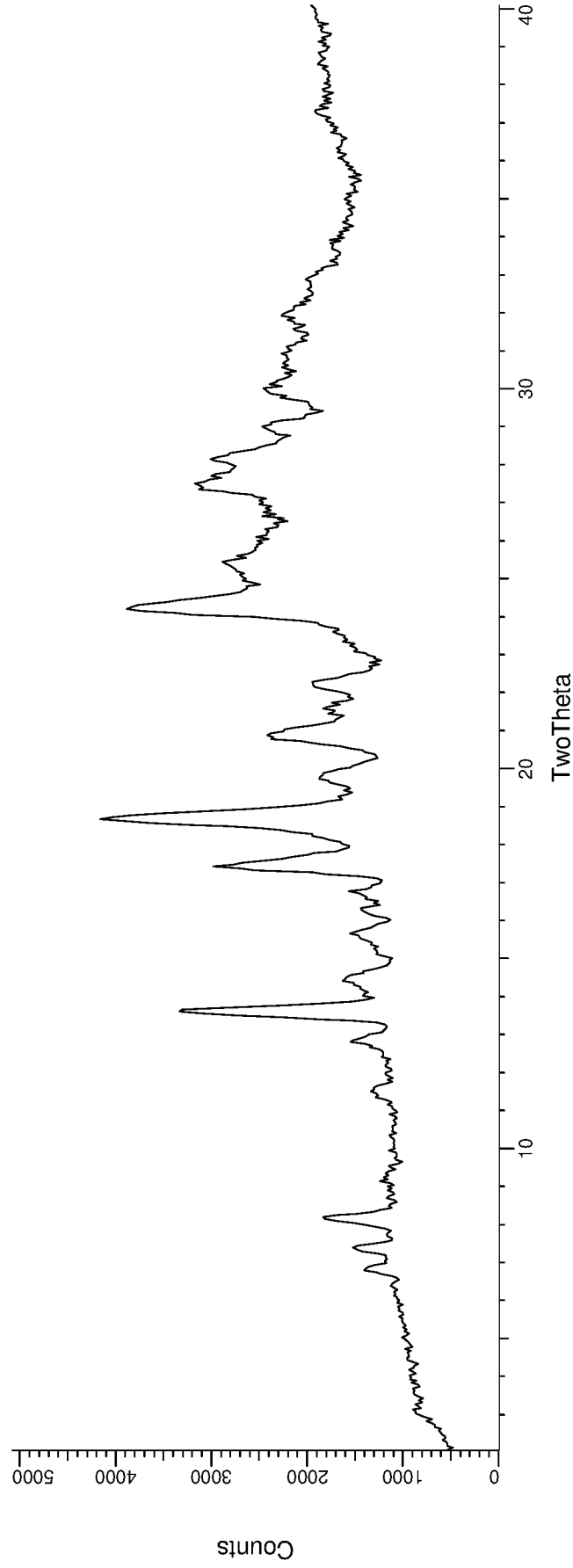


Figure 4: A characteristic XRPD of crystalline Lotilaner Form LT3

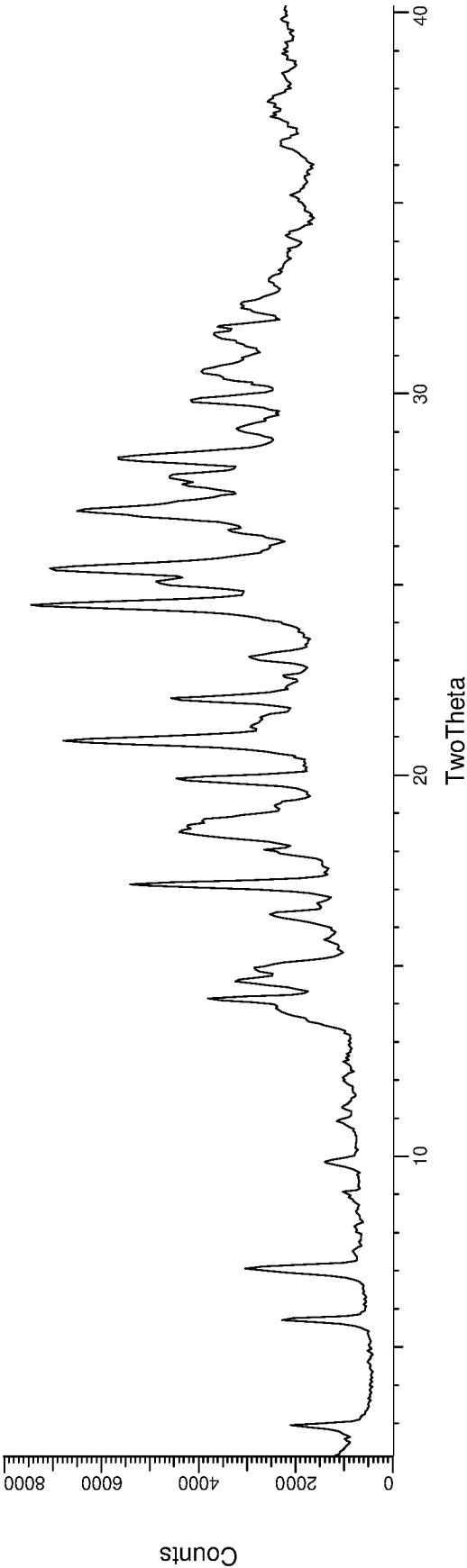


Figure 12: A characteristic TGA thermogram of crystalline Lotilaner Form LT5

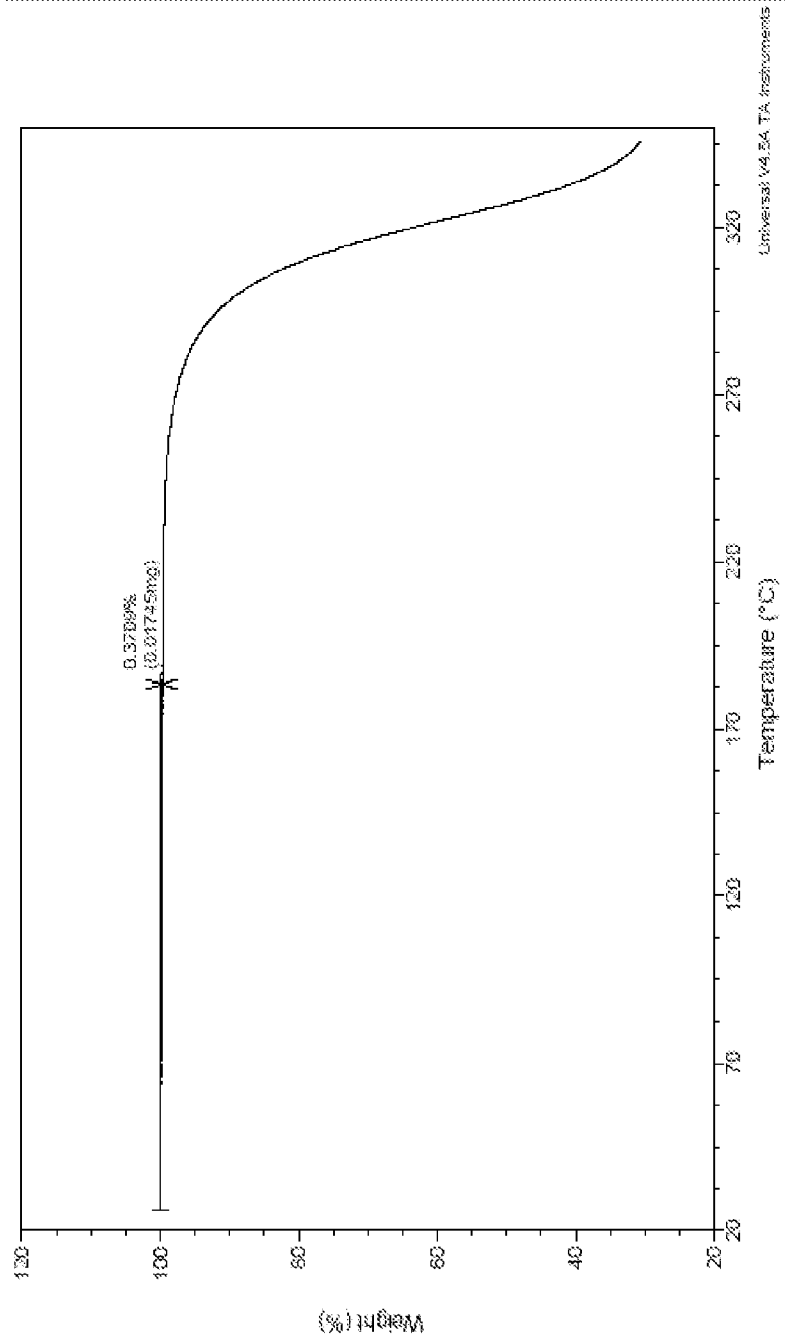
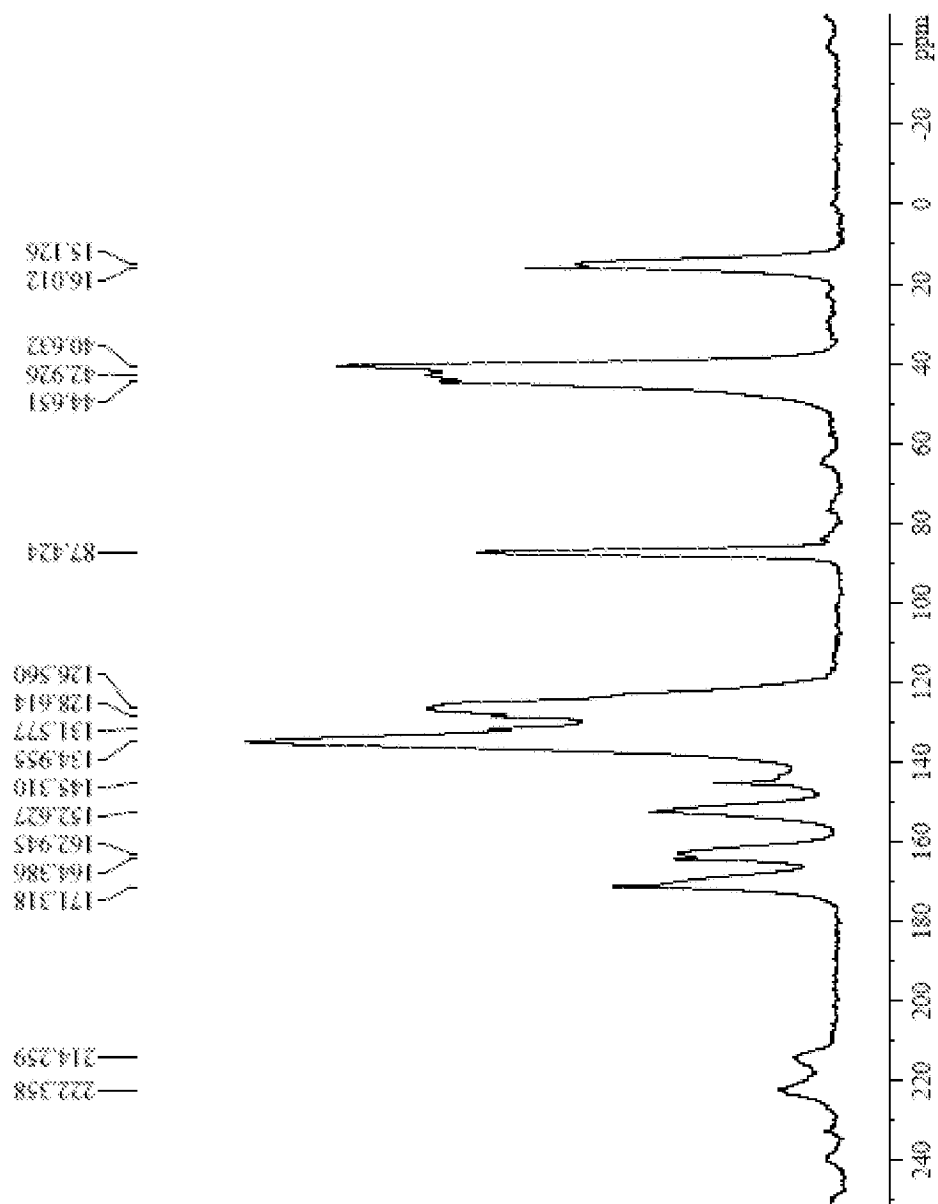


Figure 15a: A characteristic solid-state ^{13}C NMR spectrum of crystalline Lotilaner Form LT4, full scan



INTERNATIONAL SEARCH REPORT

International application No
PCT/US2022/052335

A. CLASSIFICATION OF SUBJECT MATTER INV. C07D413/04 A61K31/422 A61P27/00 A61P31/00 ADD.		
According to International Patent Classification (IPC) or to both national classification and IPC		
B. FIELDS SEARCHED Minimum documentation searched (classification system followed by classification symbols) C07D		
Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched		
Electronic data base consulted during the international search (name of data base and, where practicable, search terms used) EPO-Internal, WPI Data, BIOSIS, CHEM ABS Data		
C. DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	CN 109 879 826 A (LIVZON NEW NORTH RIVER PHARMACEUTICAL CO LTD) 14 June 2019 (2019-06-14) "The crude product was recrystallized from ethyl acetate/petroleum ether to give 88.3 g of compound 1-3, HPLC purity 99.5%, yield 93.4%." (paragraph 212); paragraph [0208] - paragraph [0213] -----	1-53
X	CN 112 457 267 A (MADISON BIOMEDICAL TECH CHENGDU CO LTD) 9 March 2021 (2021-03-09) "...recrystallize from the mixture of ethanol and water (1:2), ... to obtain the white powder product lotilaner 1-3, (10.06g, yield 72%, purity 99.64 %)" ; paragraph [0036] - paragraph [0038] ----- --/--	1-53
☒ Further documents are listed in the continuation of Box C. ☒ See patent family annex.		
* Special categories of cited documents : "A" document defining the general state of the art which is not considered to be of particular relevance "E" earlier application or patent but published on or after the international filing date "L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) "O" document referring to an oral disclosure, use, exhibition or other means "P" document published prior to the international filing date but later than the priority date claimed "T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention "X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone "Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art "&" document member of the same patent family		
Date of the actual completion of the international search		Date of mailing of the international search report
7 February 2023		15/02/2023
Name and mailing address of the ISA/ European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016		Authorized officer Stroeter, Thomas

INTERNATIONAL SEARCH REPORT

International application No

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C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2014/090918 A1 (NOVARTIS AG [CH]; NANCHEN STEVE [CH]) 19 June 2014 (2014-06-19) page 19, paragraphs 2, 3; claim 15; example 4	1-53
A	----- WO 2010/070068 A2 (NOVARTIS AG [CH]; NANCHEN STEVE [CH] ET AL.) 24 June 2010 (2010-06-24) table 2 (page 54); compound 2.4 (foam) -----	1-53

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/US2022/052335

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
CN 109879826	A	14-06-2019	NONE
CN 112457267	A	09-03-2021	NONE
WO 2014090918	A1	19-06-2014	NONE
WO 2010070068	A2	24-06-2010	AR 074790 A1 09-02-2011
		AU 2009327079 A1	24-06-2010
		BR PI0923009 A2	11-08-2015
		CA 2747354 A1	24-06-2010
		CL 2011001390 A1	28-10-2011
		CN 102256971 A	23-11-2011
		CO 6341475 A2	21-11-2011
		DK 2379537 T3	17-12-2012
		EP 2379537 A2	26-10-2011
		ES 2395704 T3	14-02-2013
		HU S1700034 I1	28-09-2017
		JP 5608676 B2	15-10-2014
		JP 2012512838 A	07-06-2012
		LU C00030 I1	10-08-2017
		NO 2017048 I1	08-09-2017
		NZ 593135 A	26-04-2013
		PL 2379537 T3	29-03-2013
		PT 2379537 E	27-12-2012
		TW 201024289 A	01-07-2010
		US 2011245239 A1	06-10-2011
		US 2013131053 A1	23-05-2013
		US 2014120147 A1	01-05-2014
		WO 2010070068 A2	24-06-2010
		ZA 201103780 B	25-01-2012